

OVERALL PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

June 9-10, 2026

June 9-10, 2026

08:00 – 09:00 Registration Open

Tuesday, June 9, 2026			Wednesday, June 10, 2026			
Time	Program (Parallel)		Time	Program (Parallel)		
9:00 – 10:30	Opening Ceremony Join with TNChE Asia 2026	TIChE Senior Project Contest 2026 (Poster Competition)	9:00 – 12:00	TIChE2026 Oral Presentation (Session 13-23)	TIChE2026 High School Project Contest (Poster Competition)	
10:30 – 12:00	Plenary Lecture Join with TNChE Asia 2026					
12:00 – 13:00	Lunch		12:00 – 13:00	Lunch		
13.00-16.30	TIChE2026 Oral Presentation (Session 1-12)	TIChE Senior Project Contest 2026 (Final Pitching)	13.00-14.30	TIChE2026 Oral Presentation (Session 25, 26)	TIChE2026 Poster Presentation	Meeting of Head of Departments
			14:30 – 15:30	Meeting on Process Simulation Software for Academic Use	Topic: Enabling a Growth Mindset and Learning at IChEmE (ONLINE)	TIChE Plant Design Competition 2026
			15:30 – 16:00	No Event	No Event	
			16:00 – 16:15	Award Ceremony for Honeywell Simulation Contest 2026		
16:30 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering Join with TNChE Asia 2026		16:15 – 17:00	Award and Closing Ceremony		

Important Notes for All Participants

- All participants for the TIChE2026 are welcome to attend all conference sessions, including joint sessions with TNChE Asia 2026, such as the Opening Ceremony, Plenary Lecture, and Networking & Exhibition Gathering.

INFORMATION FOR PRESENTERS

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

June 9-10, 2026

Presentation Code (Abstract Code)

For all participants, including invited speakers, senior project presenters, high school project contestants, and those giving poster or oral presentations, you can obtain your presentation code from your registered account on the TIChE2026.

TIChE-IV-xx	Invited Speaker
TIChE-AM-xx	Presenter in Advanced Materials and Nanotechnology (AM) Session
TIChE-BE-xx	Presenter in Biochemical and Environmental Engineering (BE) Session
TIChE-CR-xx	Presenter in Catalysis and Reaction Engineering (CR) Session
TIChE-EF-xx	Presenter in Energy and Fuels (EF) Session
TIChE-SG-xx	Presenter in Sustainable and Green Chemistry (SG) Session
TIChE-IA-xx	Presenter in Industrial Applications and Case Studies (IA) Session
TIChE-PE-xx	Presenter in Process Engineering and Digital Technologies (PE) Session
TIChE-SE-Axx	TIChE Senior Project Contest 2026 in Applied Research Category
TIChE-SE-Bxx	TIChE Senior Project Contest 2026 in Basic Research Category
TIChE-HS-xx	TIChE High School Project Contest 2026

Instruction for Oral Presentation

For all oral presentation speakers, including invited speakers and general presenters, please be informed that sessions will take place in rooms **Dusit 4, 5, 6, 7, 11, 12 and 13, 5th floor of the Dusit Thani Pattaya Hotel**. There will be a total of 26 sessions, with 5 presenters per session.

- **Invited Speakers** are allocated 30 minutes, including time for Q&A.
- **General Presenters** are allocated 10 minutes for presentation and 5 minutes for Q&A.

Please ensure that your presentation file is installed on the session computer before your session begins.

Instruction for Poster Presentation

For all poster presenters, the poster session will take place on **June 10, 2026, from 13:00-15:00, at the Activity Hall, 5th floor of the Dusit Thani Pattaya Hotel**. Presenters are kindly requested to install their posters between 12:00-12:45 prior to the session. Poster panels and necessary mounting materials will be provided by the organizing committee. Please ensure that your poster is mounted at the designated panel displaying your assigned presentation code (Abstract Code).

TECHNICAL PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : June 9, 2026

08:00 – 09:00	Registration				📍 Registration Area (5 th Floor)	
09:00 – 12:00	TICHe Senior Project Contest 2026 (Parallel) Opening Ceremony & Poster Competition				📍 Dusit 11 & Activity Hall (5 th Floor)	
09:00 – 11:00	Opening Ceremony (Join with TNChE Asia 2026)				📍 Napalai D, E (4 th Floor)	
09:00 – 10:30	Welcome Speech by Mr. Surachate Chalothorn, President of the Thai Institute of Chemical Engineering and Applied Chemistry (TICHe) Opening Ceremony of TNChE Asia 2026, TICHe2026 & Guided tour of exhibition booths					
10:30 – 11:00	Networking Break (Coffee Break)				📍 Refreshment Zone (5 th Floor)	
11:00 – 12:00	Plenary Session (Join with TNChE Asia 2026) Session Chair: Assoc. Prof. Dr. Thumrongrut Mungcharoen				📍 Napalai D, E (4 th Floor)	
11:00 – 11:20	Plenary Lecture I	AI as the New Performance Divide in the Chemicals Industry <u>Mr. Rachit Agarwal</u> Boston Consulting Group (BCG), Singapore				
11:20 – 11:40	Plenary Lecture II	Accelerating Industrial Decarbonization under Thailand's Net Zero Agenda <u>Ms. Insa Illgen</u> Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), Germany				
11:40 – 12:00	Plenary Lecture III	Financial Innovation: Unlocking Thailand's Potentials <u>Mr. Burin Adulwattana</u> Managing Director and Chief Economist, Kasikorn Research Center (KResearch), Kasikornbank, Thailand				
12:00 – 13:00	Lunch				📍 Dusit 1, 2, 3 (5 th Floor)	
13:00 – 16:30	TICHe Senior Project Contest 2026 (Parallel) Final Pitching & Award Ceremony				📍 Dusit 11 & Activity Hall (5 th Floor)	
13:00 – 14:15	TICHe2026 Oral Presentation (Parallel)				📍 Dusit 4, 5, 6, 7, 12, 13 (5 th Floor)	
Room	Dusit 4	Dusit 5	Dusit 6	Dusit 7	Dusit 12	Dusit 13
Session	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6
13:00 – 13:15	APCCChE2027	Invited Speaker TICHe-IV-08	Invited Speaker TICHe-IV-02	TICHe-EF-05	TICHe-SG-05	TICHe-PE-01
13:15 – 13:30	Invited Speaker TICHe-IV-03			TICHe-EF-06	TICHe-SG-09	TICHe-PE-02
13:30 – 13:45		TICHe-BE-03	TICHe-CR-01	TICHe-EF-07	TICHe-SG-11	TICHe-PE-03
13:45 – 14:00	Invited Speaker TICHe-IV-01	TICHe-BE-05	TICHe-CR-06	TICHe-EF-08	TICHe-SG-17	TICHe-PE-05
14:00 – 14:15		TICHe-BE-07	TICHe-CR-09	TICHe-EF-09	No Event	TICHe-PE-08
14:15 – 14:30	Networking Break (Coffee Break)				📍 Refreshment Zone (5 th Floor)	

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14:30 – 15:45	TICHe2026 Oral Presentation (Parallel)				📍 Dusit 4, 5, 6, 7, 12, 13 (5 th Floor)	
Room	Dusit 4	Dusit 5	Dusit 6	Dusit 7	Dusit 12	Dusit 13
Session	Session 7	Session 8	Session 9	Session 10	Session 11	Session 12
14:30 – 14:45	Invited Speaker TICHe-IV-06	TICHe-BE-10	TICHe-CR-19	TICHe-EF-12	TICHe-SG-19	Invited Speaker TICHe-IV-09
14:45 – 15:00		TICHe-BE-12	TICHe-CR-22	TICHe-EF-14	TICHe-SG-24	TICHe-IA-01
15:00 – 15:15	Invited Speaker TICHe-IV-05	TICHe-BE-13	TICHe-CR-11	TICHe-EF-16	TICHe-SG-25	TICHe-IA-02
15:15 – 15:30		TICHe-BE-14	TICHe-CR-12	TICHe-EF-11	TICHe-SG-26	TICHe-IA-03
15:30 – 15:45	No Event	TICHe-BE-21	TICHe-CR-16	No Event	TICHe-SG-31	No Event
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)				📍 Napalai A, B, C, D, E (4 th Floor)	
END OF DAY 1						

TECHNICAL PROGRAM**The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)****Day 2 : June 10, 2026**

08:00 – 09:00	Registration					📍 Registration Area (5 th Floor)
09:00 – 12:00	TIChE High School Project Contest 2026 (Parallel) Opening Ceremony, Poster Competition & Award Ceremony					📍 Activity Hall (5 th Floor)
09:00 – 10:15	TIChE2026 Oral Presentation (Parallel)					📍 Dusit 4, 5, 6, 7, 11, 12 & 13 (5 th Floor)
Room	Dusit 4	Dusit 5	Dusit 6	Dusit 7	Dusit 11	Dusit 12 & 13
Session	Session 13	Session 14	Session 15	Session 16	Session 17	Session 18
09:00 – 09:15	TIChE-AM-01	TIChE-BE-15	Invited Speaker TIChE-IV-07	TIChE-PE-09	TIChE-IA-04	Invited Speaker TIChE-IV-04
09:15 – 09:30	TIChE-AM-04	TIChE-BE-18		TIChE-PE-10	TIChE-IA-07	
09:30 – 09:45	TIChE-AM-08	TIChE-BE-20	TIChE-CR-21	TIChE-PE-12	TIChE-IA-09	TIChE-SG-08
09:45 – 10:00	TIChE-AM-09	TIChE-BE-08	TIChE-CR-20	TIChE-PE-13	TIChE-IA-10	TIChE-SG-36
10:00 – 10:15	No Event	No Event	TIChE-CR-24	TIChE-PE-14	TIChE-PE-04	TIChE-SG-37
10:15 – 10:30	Networking Break (Coffee Break)					📍 Refreshment Zone (5 th Floor)

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10:30 – 11:45	TIChE2026 Oral Presentation (Parallel)					Dusit 4, 5, 6, 7, 11, 12&13 (5 th Floor)
Room	Dusit 4	Dusit 5	Dusit 6	Dusit 7	Dusit 11	Dusit 12 & 13
Session	Session 19	Session 20	Session 21	Session 22	Session 23	Session 24
10:30 – 10:45	TIChE-AM-10	TIChE-PE-06	TIChE-CR-26	TIChE-AM-19	TIChE-SG-39	No Event
10:45 – 11:00	TIChE-AM-11	TIChE-PE-17	TIChE-CR-27	TIChE-AM-20	TIChE-SG-40	No Event
11:00 – 11:15	TIChE-AM-13	TIChE-PE-19	TIChE-CR-32	TIChE-AM-21	TIChE-SG-43	No Event
11:15 – 11:30	TIChE-AM-14	TIChE-PE-20	TIChE-CR-34	TIChE-AM-23	TIChE-SG-06	No Event
11:30 – 11:45	TIChE-AM-15	TIChE-PE-21	TIChE-CR-37	TIChE-AM-24	TIChE-SG-33	No Event
11:45 – 13:00	Lunch					Dusit 1, 2, 3 (5 th Floor)
13:00 – 15:00	TIChE2026 Poster Presentation (Parallel)					Activity Hall (5 th Floor)
13:00 – 14:15	TIChE2026 Oral Presentation (Parallel) & Meeting					Dusit 4, 5, 6, 7, 11, 12&13 (5 th Floor)
Room	Dusit 4	Dusit 5	Dusit 6	Dusit 7	Dusit 11	Dusit 12 & 13
Session	Session 25	Session 26	Session 27	Session 28	Session 29	Session 30
13:00 – 13:15	 TIChE-PE-22	TIChE-CR-14	No Event	No Event	No Event	Meeting of the Heads of Chemical Engineering Departments of Thailand
13:15 – 13:30	TIChE-AM-25	TIChE-CR-07	No Event	No Event	No Event	
13:30 – 13:45	TIChE-AM-26	TIChE-CR-30	No Event	No Event	No Event	
13:45 – 14:00	No Event	TIChE-CR-33	No Event	No Event	No Event	
14:00 – 14:15	No Event	No Event	No Event	No Event	No Event	
14:15 – 14:30	Networking Break (Coffee Break)					Refreshment Zone (5 th Floor)

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14:30 – 16:15	TIChE Plant Design Competition 2026, Award & Meeting				📍 Dusit 6 & 7, 11, 12&13 (5 th Floor)		
Room	Dusit 4	Dusit 5	Dusit 6 & 7		Dusit 11	Dusit 12 & 13	
Session	No Event	No Event	Session 34	Session 33	Session 32	Session 31	
14:30 – 15:30	No Event	No Event	Topic: Enabling a Growth Mindset and Learning at IChemE (ONLINE)  TIChE-IV-10		Meeting on Process Simulation Software for Academic Use	TIChE Plant Design Competition 2026	
15:30 – 16:00							
16:00 – 16:15			No Event		No Event	No Event	Award Ceremony for the TIChE-Honeywell Simulation Contest 2026
16:15 – 17:00							Award and Closing Ceremony
END OF DAY 2							
END OF TIChE2026 CONFERENCE							

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 1 : Advanced Materials and Nanotechnology (AM)

📍 Dusit 4 (5th Floor)
🕒 13:00 – 14:15

Chair : Asst. Prof. Dr. Idtisak Paopo

Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand

Time	Abstract Code	Title	Presenter
13:00 – 13:15	APCChE2027	Introduction to APCChE 2027	 Prof. Dr. Hui Tong Chua Department of Chemical Engineering, School of Engineering, The University of Western Australia
13:15 – 13:45	Invited Speaker TIChE-IV-03	Lamp Ablation Technology for Accessing Various Nanostructures	
13:45 – 14:15	Invited Speaker TIChE-IV-01	Natural Rubber-Based Sustainable Packaging: 10 Years of Innovation from Functional Performance to End-of-Life Degradation	 Prof. Dr. Cattaleeya Pattamaprom Department of Chemical Engineering, Thammasat School of Engineering, Thammasat University, Thailand
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

Session 2 : Biochemical and Environmental Engineering (BE)

📍 Dusit 5 (5th Floor)
🕒 13:00 – 14:15

Chair : Assoc. Prof. Dr. Sommai Pivsa-Art

Department of Materials and Metallurgical Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Thailand

Co-Chair : Asst. Prof. Dr. Kantharakorn Macharoen

Department of Chemical Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi, Thailand

Time	Abstract Code	Title	Presenter
13:00 – 13:15	Invited Speaker TIChE-IV-08	Unlocking the Potential of Metal-Organic Frameworks (MOFs) for Water Purification Strategies	 Assoc. Prof. Dr. Thanh-Binh Nguyen College of Hydrosphere Science, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan
13:15 – 13:30			
13:30 – 13:45	TIChE-BE-03	ANN-LSTM hybrid model for Thailand's MSW and GHG prediction 2050	Ms. Pichchaporn Tongseelaung Thammasat University, Thailand
13:45 – 14:00	TIChE-BE-05	Enhanced removal of methylene blue and methyl red using cationic surfactant modified sodium and calcium bentonite composite with tailored rice husk ash	Mr. Asadullah Suranaree University of Technology, Thailand
14:00 – 14:15	TIChE-BE-07	Efficient removal of turbidity and organic matter from wet coffee processing wastewater using coagulation–flocculation	Ms. Jitapa Muangsiri King Mongkut's University of Technology Thonburi, Thailand
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 3 : Catalysis and Reaction Engineering (CR)			📍 Dusit 6 (5 th Floor) 🕒 13:00 – 14:15
Chair : Prof. Dr. Thongthai Wittoon Department of Chemical Engineering, Faculty of Engineering, Kasetsart University, Thailand Co-Chair : Asst. Prof. Dr. Asad Ullah School of Chemical Engineering, Institute of Engineering, Suranaree University of Technology, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:30	Invited Speaker TIChE-IV-02	Glass Engineering of Metal-Organic Frameworks/Coordination Polymers for Composite Catalytic Systems	 Assoc. Prof. Dr. Hiroyasu Tabe Institute for Integrated Cell-Material Sciences, Kyoto University, Japan
13:30 – 13:45	TIChE-CR-01	Solid-solid thermal method as green alternative method to synthesize MOFs and their derivative materials	Assoc. Prof. Dr. Somboon Chaemchuen Mahidol University, Thailand
13:45 – 14:00	TIChE-CR-06	Development of Pd/Al ₂ O ₃ -based catalytic converter for gasoline-ammonia dual-fuel engine	Ms. Pimchanok Phuthongking Chulalongkorn University, Thailand
14:00 – 14:15	TIChE-CR-09	Bio-naphtha production from palm oil hydroprocessing over IrRe-based catalysts: Effect of third metal incorporation	Ms. Nichakorn Taejajai Chulalongkorn University, Thailand
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

Session 4 : Energy and Fuels (EF)			📍 Dusit 7 (5 th Floor) 🕒 13:00 – 14:15
Chair : Prof. Dr. Pornpote Piumsomboon Department of Chemical Technology, Faculty of Science, Chulalongkorn University, Thailand Co-Chair : Assoc. Prof. Dr. Sukritthira Ratanawilai Department of Chemical Engineering, Faculty of Engineering, Prince of Songkla University, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:15	TIChE-EF-05	Degree of crosslinking optimization for quaternized polyvinyl alcohol as anion exchange membranes in water electrolysis	Ms. Panita Insang Chulalongkorn University, Thailand
13:15 – 13:30	TIChE-EF-06	Techno-economic feasibility of upgrading biogas plants for hydrogen production via dry reforming of methane (DRM): A case study for Thailand	Mr. Teetat Wattanasilp Thammasat University, Thailand
13:30 – 13:45	TIChE-EF-07	Factorial design optimization of temperature and WHSV for maximizing light biofuel and biokerosene from waste cooking oil	Mr. Kasidit Loplamlert Khon Kaen University, Thailand
13:45 – 14:00	TIChE-EF-08	Catalytic pyrolysis of waste cooking oil to aviation-range biofuels in a continuous packed-bed reactor	Mr. Thitikorn Inthakhinee Khon Kaen University, Thailand
14:00 – 14:15	TIChE-EF-09	Life cycle assessment of sustainable aviation fuel from oil palm and palm oil industry residues	Mr. Nontabhun Chaibun King Mongkut's University of Technology Thonburi, Thailand
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 5 : Sustainable and Green Chemistry (SG)			📍 Dusit 12 (5 th Floor) 🕒 13:00 – 14:15
Chair : Prof. Dr. Somkiat Ngamprasertsith Department of Chemical Technology, Faculty of Science, Chulalongkorn University, Thailand Co-Chair : Asst. Prof. Dr. Prodpran Siritheerasas Department of Chemical Engineering, School of Engineering, Thammasat University, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:15	TIChE-SG-05	Enhancing cellulose nanocrystal production: A novel Lewis acid approach with Iron (III) chloride	Ms. Rinthara Witayanuphap Mahidol University, Thailand
13:15 – 13:30	TIChE-SG-09	Circular concept for spent LIB from E-motorbikes	Mr. Parinya Nunoi Thammasat University, Thailand
13:30 – 13:45	TIChE-SG-11	Effect of Ca(OH) ₂ dissolution rate on CaCO ₃ polymorph and particle size in CO ₂ -loaded MEA	Mr. Sirodom Tangprakon Chulalongkorn University, Thailand
13:45 – 14:00	TIChE-SG-17	Monoglyceride synthesis via glycerolysis of oleic acid in an oscillatory baffled reactor	Mr. Thanathorn Wangsalaeh Chulalongkorn University, Thailand
14:00 – 14:15	No Event		
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

Session 6 : Process Engineering and Digital Technologies (PE)			📍 Dusit 13 (5 th Floor) 🕒 13:00 – 14:15
Chair : Assoc. Prof. Dr. Yaneeporn Patcharavorachot Department of Chemical Engineering, Faculty of Engineering, King Mongkut's Institute of Technology Ladkrabang, Thailand Co-Chair : Assoc. Prof. Dr. Kulchanat Prasertsit Department of Chemical Engineering, Faculty of Engineering, Prince of Songkla University, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:15	TIChE-PE-01	Digital-twin-oriented pedagogical integration of BatchSep for dynamic modeling and validation of MeOH/H ₂ O batch distillation	Mr. Anawin Ampaipisut King Mongkut's University of Technology North Bangkok, Thailand
13:15 – 13:30	TIChE-PE-02	Techno-economic analysis of diethyl carbonate production via oxidative carbonylation of bioethanol	Ms. Suthinee Saetang Chulalongkorn University, Thailand
13:30 – 13:45	TIChE-PE-03	Hybrid-AI-sep: An AI-augmented computer-aided tool for separation process design	Mr. Kris Prasopsanti PSE for SPEED Company Ltd., Bangkok, Thailand
13:45 – 14:00	TIChE-PE-05	Hybrid-AI-cafd: Integrating process systems engineering and AI for intelligent process synthesis, design, & innovation	Dr. Rungroj Yadbantung PSE for SPEED Company Ltd., Bangkok, Thailand
14:00 – 14:15	TIChE-PE-08	Performance mapping of Thai agricultural residues for cogeneration: A rapid operational decision-support across diverse feedstock conditions	Mr. Kittipon Jaraswimol Kasetsart University, Thailand
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 7 : Advanced Materials and Nanotechnology (AM)

📍 Dusit 4 (5th Floor)
🕒 14:30 – 15:45

Chair : Asst. Prof. Dr. Idtisak Paopo

Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand

Time	Abstract Code	Title	Presenter
14:30 – 15:00	Invited Speaker TIChE-IV-06	Carbohydrate Based Biomaterials for Biomedical Applications	 Prof. Dr. Ravin Narain College of Natural and Applied Sciences, University of Alberta, Canada
15:00 – 15:30	Invited Speaker TIChE-IV-05	Functional Biopolymers Integrated with Bioactive Compounds	 Prof. Dr. Pornnapa Kasemsiri Department of Chemical Engineering, Faculty of Engineering, Khon Kaen University, Thailand
15:30 – 15:45	No Event		
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

Session 8 : Biochemical and Environmental Engineering (BE)

📍 Dusit 5 (5th Floor)
🕒 14:30 – 15:45

Chair : Assoc. Prof. Dr. Asawin Meechai

Department of Chemical Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi, Thailand

Co-Chair : Asst. Prof. Dr. Emma Asnachinda

Department of Chemical Engineering, Burapha University, Chonburi, Thailand

Time	Abstract Code	Title	Presenter
14:30 – 14:45	TIChE-BE-10	Membrane configuration effects on flux stability and fouling in an aerated membrane bioreactor treating rubber processing wastewater	Mr. Satakhun Imsomboon King Mongkut's University of Technology Thonburi, Thailand
14:45 – 15:00	TIChE-BE-12	Photocatalytic degradation of rhodamine B using K-doped biochar/graphitic carbon nitride composite	Ms. Chheng Im Si Chulalongkorn University, Thailand
15:00 – 15:15	TIChE-BE-13	Structural modification of hydroxyapatite derived from eggshell using gamma radiation and iron dopant for wastewater remediation	Ms. Geovanna Asitimbay Chulalongkorn University, Thailand
15:15 – 15:30	TIChE-BE-14	Microwave-assisted organosolv extraction of lignin from sugarcane bagasse using glycerol and organic acids	Mr. Chonlathit Chitpisitchai King Mongkut's University of Technology Thonburi, Thailand
15:30 – 15:45	TIChE-BE-21	Adsorption kinetics and isotherms of the removal of food colouring using various types of activated carbon	Asst. Prof. Dr. Prodpran Siritheerasas Thammasat University, Thailand
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 9 : Catalysis and Reaction Engineering (CR)			📍 Dusit 6 (5 th Floor) 🕒 14:30 – 15:45
Chair : Assoc. Prof. Dr. Somboon Chaemchuen Department of Chemical Engineering, Faculty of Engineering, Mahidol University, Thailand Co-Chair : Assoc. Prof. Dr. Waleeporn Donphai Department of Chemical Engineering, Faculty of Engineering, Kasetsart University, Thailand			
Time	Abstract Code	Title	Presenter
14:30 – 14:45	TIChE-CR-19	Synthesis of biodiesel from waste cooking oil using interesterification process	Mr. Kittipong Klinmontha King Mongkut's Institute of Technology Ladkrabang, Thailand
14:45 – 15:00	TIChE-CR-22	Catalyst selection and condition optimization for synthesis of carbon nanotubes from biomass and plastic waste mixture	Ms. Thanchanok Vettavong Chulalongkorn University, Thailand
15:00 – 15:15	TIChE-CR-11	Influence of catalyst support on reaction selectivity in CO ₂ methanation over Ni catalysts	Ms. Kamonrat Suksumrit Thailand Institute of Scientific and Technological Research (TISTR), Thailand
15:15 – 15:30	TIChE-CR-12	Glycerol acetylation with acetic acid catalyzed by ion-exchange resin-derived nanoporous carbon functionalized via hydrothermal sulfonation with sulfuric acid and Lek Namphi	Mr. Krit Nopparee Chulalongkorn University, Thailand
15:30 – 15:45	TIChE-CR-16	Production of monolaurin from glycerol via esterification of lauric acid over a ZnO/γ-Al ₂ O ₃ catalyst	Ms. Chalisa Boonmee Rajamangala University of Technology Thanyaburi, Thailand
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

Session 10 : Energy and Fuels (EF)			📍 Dusit 7 (5 th Floor) 🕒 14:30 – 15:30
Chair : Prof. Dr. Kitirote Wantala Department of Chemical Engineering, Faculty of Engineering, Khon Kaen University, Thailand Co-Chair : To be announced			
Time	Abstract Code	Title	Presenter
14:30 – 14:45	TIChE-EF-12	Effects of operating conditions on green hydrogen generation via proton exchange membrane water electrolysis	Mr. Sakada Poev Chulalongkorn University, Thailand
14:45 – 15:00	TIChE-EF-14	Polarization behavior of an alkaline water electrolysis at different current density and temperature	Ms. Kimlay Ngorn Chulalongkorn University, Thailand
15:00 – 15:15	TIChE-EF-16	Techno-economic analysis and life cycle assessment of sustainable aviation fuel production from used cooking oil	Mr. Sarun Lertsinsubtwee Kasetsart University, Thailand
15:15 – 15:30	TIChE-EF-11	Techno-economic evaluation of green hydrogen production: Impacts on levelized cost of hydrogen (LCOH)	Ms. Kanokwan Maikaew National Science and Technology Development Agency (NSTDA), Thailand
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 1 : Tuesday, June 9, 2026

Session 11 : Sustainable and Green Chemistry (SG)

📍 **Dusit 12 (5th Floor)**
🕒 **14:30 – 15:45**

Chair : **Assoc. Prof. Dr. Sommai Pivsa-Art**

Department of Materials and Metallurgical Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Thailand

Co-Chair : **Assoc. Prof. Dr. Weraporn Pivsa-Art**

Department of Chemical and Materials Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Thailand

Time	Abstract Code	Title	Presenter
14:30 – 14:45	TIChE-SG-19	Monoglycerides production from palm kernel oil via glycerolysis in a mechanically stirred reactor	Mr. Pattarakorn Yensuang Rajamangala University of Technology Thanyaburi, Thailand
14:45 – 15:00	TIChE-SG-24	Using a self-fabricated magnetite containing bio-based polymer as a heterogeneous catalyst electrode in Electro-Fenton Process	Ms. Natchaya Atibaedya Mahidol University, Thailand
15:00 – 15:15	TIChE-SG-25	Comparative LCA of amine absorption and cryogenic carbon capture from natural gas power plant flue gas: A case study in Thailand	Mr. Pakapon Kitipaisalnont Kasetsart University, Thailand
15:15 – 15:30	TIChE-SG-26	Utilization of biomass-derived biochar as a support material for development of renewable solid sorbents used in post-combustion CO ₂ capture	Mr. Pongsarun Boribonwach Mahidol University, Thailand
15:30 – 15:45	TIChE-SG-31	Utilizing air-cathode biofilm by polarity reversal and low-cost hydrochar electrode modifier for hydrogen generation in microbial electrolysis cell	Mr. Hamad Kaiser Prince of Songkla University, Thailand
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

Session 12 : Industrial Applications and Case Studies (IA)

📍 **Dusit 13 (5th Floor)**
🕒 **14:30 – 15:30**

Chair : **Dr. Wasu Chaitree**

Department of Chemical Engineering, Faculty of Engineering and Industrial Technology, Silpakorn University, Thailand

Co-Chair : **Assoc. Prof. Dr. Chatchai Kunyawut**

Department of Chemical and Environmental Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand

Time	Abstract Code	Title	Presenter
14:30 – 14:45	Invited Speaker TIChE-IV-09	A Comparison of Ground and Air Source Heat Pump Performance for Domestic Applications: A Case Study in Perth, Australia	 Prof. Dr. Hui Tong Chua Department of Chemical Engineering, School of Engineering, The University of Western Australia
14:45 – 15:00	TIChE-IA-01	Long-term field exposure performance of one-part geopolymer mortars in a live sewer environment	Asst. Prof. Dr. Cherdphong Seedao Burapha University, Thailand
15:00 – 15:15	TIChE-IA-02	Impact of the storage time on the bioactive compounds of the freeze-dried fine robusta coffee	Mr. Pongsapak Rattanachai King Mongkut's University of Technology Thonburi, Thailand
15:15 – 15:30	TIChE-IA-03	Data-driven predictive model for aluminium anodizing process using machine learning technique	Miss Natthanant Tanthatemee Silpakorn University, Thailand
15:45 – 18:30	Networking & Happy Hour at Exhibition Zone Gathering (Join with TNChE Asia 2026)		📍 Napalai A, B, C, D, E (4 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 13 : Advanced Materials and Nanotechnology (AM)

📍 Dusit 4 (5th Floor)
🕒 09:00 – 10:00

Chair : **Assoc. Prof. Dr. Weerinda Mens**

Department of Chemical and Materials Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Thailand

Co-Chair : **Assoc. Prof. Dr. Weraporn Pivsa-Art**

Department of Chemical and Materials Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Thailand

Time	Abstract Code	Title	Presenter
09:00 – 09:15	TIChE-AM-01	Polyethylenimine-filled silica and layered double hydroxides network for efficient removal of tetracycline	Miss Windi Agustina Chulalongkorn University, Thailand
09:15 – 09:30	TIChE-AM-04	Upcycled Carbon Rods to Defective Graphite as Sodium-Insertion and Deposition Electrodes for Sodium-ion Batteries	Mr. Hafs Musa Abubakar Chulalongkorn University, Thailand
09:30 – 09:45	TIChE-AM-08	Influence of organic solvents on the characteristics of kraft lignin nanoparticles synthesized by controlled solvent shifting	Ms. Wasinee Ruengkao King Mongkut's University of Technology North Bangkok, Thailand
09:45 – 10:00	TIChE-AM-09	Optimization of electron beam irradiation parameters for upcycling residual liquid silicone rubber into mattress pads for newborn phototherapy	Ms. Chanunya Pratchayakul Chulalongkorn University, Thailand
10:00 – 10:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

Session 14 : Biochemical and Environmental Engineering (BE)

📍 Dusit 5 (5th Floor)
🕒 09:00 – 10:00

Chair : **Assoc. Prof. Dr. Lek Wantha**

School of Chemical Engineering, Institute of Engineering, Suranaree University of Technology

Co-Chair : **To be announced**

Time	Abstract Code	Title	Presenter
09:00 – 09:15	TIChE-BE-15	Comparative degradation behavior of PLA and PBS bioplastics under soil burial and UV-A radiation	Ms. Emma Asnachinda Burapha University, Thailand
09:15 – 09:30	TIChE-BE-18	Synthesis characterization and carbonfootprint of carboxymethyl cellulose derived from rice straw	Ms. Wilasinee Sangchan Thammasat University, Thailand
09:30 – 09:45	TIChE-BE-20	Removal of perfluorobutanesulfonic acid (PFBS) by biodegradable zwitterionic surfactant-assisted foam fractionation	Mr. Tengheang Hor Chulalongkorn University, Thailand
09:45 – 10:00	TIChE-BE-08	Estimation of Daily Intake of Microplastics from Bottled Drinking Water in Thailand	Ms. Chareerat Kuntol Thammasat University, Thailand
10:00 – 10:15	No Event		
10:15 – 10:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 15 : Catalysis and Reaction Engineering (CR)			📍 Dusit 6 (5 th Floor) 🕒 09:00 – 10:15
Chair : Prof. Dr. Anusorn Seubsai Department of Chemical Engineering, Faculty of Engineering, Kasetsart University, Thailand Co-Chair : Assoc. Prof. Dr. Supareak Prasertthdam Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University			
Time	Abstract Code	Title	Presenter
09:00 – 09:30	Invited Speaker TIChE-IV-07	Quantum Chemistry-informed Optimization of Heterogeneous Catalytic Processes: A Case Study of Integrated Carbon Capture and Conversion	 Assoc. Prof. Dr. Supareak Prasertthdam Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand
09:30 – 09:45	TIChE-CR-21	Process optimization of multi-walled carbon nanotubes and hydrogen co-production capacity from polyethylene in a tubular batch reactor	Mr. Warodom Thepmongkorn Chulalongkorn University, Thailand
09:45 – 10:00	TIChE-CR-20	Microwave-assisted acetylation of glycerol to triacetin over tungstophosphoric acid supported on activated carbon	Ms. Chutikan Aiemvasan Rajamangala University of Technology Thanyaburi, Thailand
10:00 – 10:15	TIChE-CR-24	Effect of preheating and sulfonation on Amberlyst 15 activity for selective 5-HMF synthesis from sucrose in continuous systems	Miss Maturada Suksaard Mahidol University, Thailand
10:15 – 10:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

Session 16 : Process Engineering and Digital Technologies (PE)			📍 Dusit 7 (5 th Floor) 🕒 09:00 – 10:15
Chair : Assoc. Prof. Dr. Chaiwat Prapainainar Department of Chemical Engineering, Faculty of Engineering, King Mongkut's University of Technology North Bangkok, Thailand Co-Chair : Dr. Suparat Sasrimuang Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand			
Time	Abstract Code	Title	Presenter
09:00 – 09:15	TIChE-PE-09	Polymorphic behaviour and solubility of L-lysine monohydrochloride for in-situ monitoring of cooling crystallisation	Mr. Lovejoy Nyoni Suranaree University of Technology, Thailand
09:15 – 09:30	TIChE-PE-10	Time-varying multivariate statistical process control for batch processes	Asst. Prof. Dr. Bundit Boonkhao Nakhon Phanom University, Thailand
09:30 – 09:45	TIChE-PE-12	Design of Model Predictive Control for Tubular Reactors with Cooling Jacket	Ms. Chonthicha Somrang Silpakorn University, Thailand
09:45 – 10:00	TIChE-PE-13	Integration of process simulation and LSTM for boiler operating phase classification toward digital twin deployment	Ms. Chularak Takhonram Mahidol University, Thailand
10:00 – 10:15	TIChE-PE-14	Improving benzoxazinone yield through temperature control using model predictive control	Mr. Jaturon Thapim Chulalongkorn University, Thailand
10:15 – 10:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 17 : Industrial Applications and Case Studies (IA) Process Engineering and Digital Technologies (PE)			 Dusit 11 (5th Floor)  09:00 – 10:15
Chair : Assoc. Prof. Dr. Chaiyan Chaiya Department of Chemical and Materials Engineering, Rajamangala University of Technology Thanyaburi, Thailand Co-Chair : Assoc. Prof. Tharathon Mongkhonsi Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University, Thailand			
Time	Abstract Code	Title	Presenter
09:00 – 09:15	TIChE-IA-04	Effect of coffee processes on bioactive compounds of the green arabica coffee beans	Ms. Theeraya Kulawong King Mongkut's University of Technology Thonburi, Thailand
09:15 – 09:30	TIChE-IA-07	Water sorption isotherm of sugar and liquid bridging point at different temperatures	Dr. Terasut Sookkumnerd Suranaree University of Technology, Thailand
09:30 – 09:45	TIChE-IA-09	Investigation of kinetic models for high-pressure liquid extraction of turmeric	Mr. Athippracha Wattanawijit Burapha University, Thailand
09:45 – 10:00	TIChE-IA-10	ChemSub: A computer-aided tool for chemical substitution	Ms. Thunyaras Phanusupawimol PSE for SPEED Company Ltd., Bangkok, Thailand
10:00 – 10:15	TIChE-PE-04	A Unified Molecular-Structure-Based Framework for Pure-Component and Mixture Property Estimation	Mr. Kris Prasopsanti PSE for SPEED Company Ltd., Bangkok, Thailand
10:15 – 10:30	Networking Break (Coffee Break)		 Refreshment Zone (5th Floor)

Session 18 : Sustainable and Green Chemistry (SG)			 Dusit 12&13 (5th Floor)  09:00 – 10:15
Chair : Assoc. Prof. Dr. Chularat Sakdaronnarong Department of Chemical Engineering, Faculty of Engineering, Mahidol University, Thailand Co-Chair : Asst. Prof. Dr. Saiwan Nawalertpanya Department of Chemical Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi, Thailand			
Time	Abstract Code	Title	Presenter
09:00 – 09:30	Invited Speaker TIChE-IV-04	Electrolyte Engineering: From Solution Thermodynamics to High-Voltage Energy Storage Applications	 Asst. Prof. Dr. Pawin lamprasertkun School of Bio-Chemical Engineering and Technology, Sirindhorn International Institute of Technology, Thammasat University, Thailand
09:30 – 09:45	TIChE-SG-08	Evaluating the leaching potential for Cu extraction from electronic scrap	Ms. Kedsara Songprasert Thammasat University, Thailand
09:45 – 10:00	TIChE-SG-36	Integrated carbon dioxide absorption and mineralization derived calcium carbonate as a carbon dioxide source in dry reforming of methane	Ms. Paweethida Nguankratok Chulalongkorn University, Thailand
10:00 – 10:15	TIChE-SG-37	Novel PET-to-TPA recycling via neutral hydrolysis over HZSM-5 zeolite catalyst	Mr. Nawantorn Jaisawang Khon Kaen University, Thailand
10:15 – 10:30	Networking Break (Coffee Break)		 Refreshment Zone (5th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 19 : Advanced Materials and Nanotechnology (AM)

📍 Dusit 4 (5th Floor)
🕒 10:30 – 11:45

Chair : Prof. Dr. Pornnapa Kasemsiri

Department of Chemical Engineering, Faculty of Engineering, Khon Kaen University, Thailand

Co-Chair : To be announced

Time	Abstract Code	Title	Presenter
10:30 – 10:45	TIChE-AM-10	Binary thermoplastic polyurethane blends for highly thermal and physical properties for cushioning applications	Ms. Chatchadaporn Koedpakpraek Chulalongkorn University, Thailand
10:45 – 11:00	TIChE-AM-11	Hand made Thai silk tensile strength quality improvement with chitam coating	Mr. Phusit Prasom King Mongkut's Institute of Technology Ladkrabang, Thailand
11:00 – 11:15	TIChE-AM-13	Enhancing the durability: An investigation of bamboo impregnation for preserving its properties	Mr. Thananan Thongyoi Chulalongkorn University, Thailand
11:15 – 11:30	TIChE-AM-14	CNT/PDMS composites used as flexible strain sensors: Influence of CNT diameter and catalyst impurities	Ms. Phitchawalai Nintachan Mahidol University, Thailand
11:30 – 11:45	TIChE-AM-15	Effects of slow pyrolysis temperature on physicochemical properties of biochar derived from Sang-Mon bamboo industrial residues	Mr. Nontakorn Chuyingsakultrip Chulalongkorn University, Thailand
11:45 – 12:00	Lunch		📍 Dusit 1, 2, 3 (5 th Floor)

Session 20 : Process Engineering and Digital Technologies (PE)

📍 Dusit 5 (5th Floor)
🕒 10:30 – 11:45

Chair : Assoc. Prof. Dr. Chatchai Kunyawut

Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand

Co-Chair : To be announced

Time	Abstract Code	Title	Presenter
10:30 – 10:45	TIChE-PE-06	Improving biobutanol production through IBE fermentation using optimization-based adaptive PID control	Mr. Watchanan Saengsuwan Burapha University, Thailand
10:45 – 11:00	TIChE-PE-17	Physical property laboratory with special focus on data for chemical process simulation and optimization	Miss Sunisa Singhawannurat Rareytec Company, Ltd., Thailand
11:00 – 11:15	TIChE-PE-19	Process simulation of cryogenic biogas upgrading integrated with liquefaction utilizing mixed refrigerants	Assoc. Prof. Dr. Anawat Sungpet King Mongkut's University of Technology Thonburi, Thailand
11:15 – 11:30	TIChE-PE-20	Machine learning for forward and inverse modeling of structure-property relationships in polypropylene in-reactor alloy	Mr. Kett Khayanying Kasetsart University, Thailand
11:30 – 11:45	TIChE-PE-21	Machine learning for forward and inverse modeling of linear low-density polyethylene	Mr. Sirawich Chomchao Kasetsart University, Thailand
11:45 – 12:00	Lunch		📍 Dusit 1, 2, 3 (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 21 : Catalysis and Reaction Engineering (CR)			9 Dusit 6 (5 th Floor) 10:30 – 11:45
Chair : Asst. Prof. Dr. Choowong Chaisuk Department of Chemical Engineering, Faculty of Engineering and Industrial Technology, Silpakorn University, Thailand Co-Chair : Asst. Prof. Dr. Idtisak Paopo Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand			
Time	Abstract Code	Title	Presenter
10:30 – 10:45	TIChE-CR-26	Roles of the ratio of TiO ₂ and ZrO ₂ in the Cu based catalyst on the selective hydrogenation of furfural to furfuryl alcohol	Ms. Sawitree Ubolwat Silpakorn University, Thailand
10:45 – 11:00	TIChE-CR-27	Promotional effect of Zr on Cu/TiO ₂ catalyst for the liquid-phase selective hydrogenation of furfural to furfuryl alcohol	Ms. Phimchanok Phoromyen Silpakorn University, Thailand
11:00 – 11:15	TIChE-CR-32	Electrooxidation of glycerol on NiMo electrocatalyst supported on carbon cloth to value-added products	Dr. Wasu Chaitree Silpakorn University, Thailand
11:30 – 11:45	TIChE-CR-34	Chemical recycling of expanded polystyrene (EPS) foam waste via photocatalytic degradation	Ms. Supitcha Suksa-ard Sirindhorn International Institute of Technology (SIIT), Thammasat University, Thailand
11:15 – 11:30	TIChE-CR-37	Effect of catalyst particle size on heat transfer and reaction performance in a packed bed reactor for CO ₂ hydrogenation using computational fluid dynamics	Ms. Nattha Charoenchasri Chulalongkorn University, Thailand
11:45 – 12:00	Lunch		9 Dusit 1, 2, 3 (5 th Floor)

Session 22 : Advanced Materials and Nanotechnology (AM)			9 Dusit 7 (5 th Floor) 10:30 – 11:45
Chair : Asst. Prof. Dr. Asad Ullah School of Chemical Engineering, Institute of Engineering, Suranaree University of Technology, Thailand Co-Chair : Dr. Kulpavee Jitapunkul Department of Chemical Engineering, Faculty of Engineering, Kasetsart University, Thailand			
Time	Abstract Code	Title	Presenter
10:30 – 10:45	TIChE-AM-19	Molecular design of silane-based self-assembled monolayers for enhancing carrier extraction in perovskite solar cells	Mr. Peradon Phonglamjiangkang Chulalongkorn University, Thailand
10:45 – 11:00	TIChE-AM-20	Microbubble processing of materials for energy storage applications	Dr. Pichamon Sirisinudomkit Prince of Songkla University, Thailand
11:00 – 11:15	TIChE-AM-21	Multilayer POM fiber-reinforced PBS structures for engineering materials	Ms. Satoko Shiroguchi Rajamangala University of Technology Thanyaburi, Thailand
11:30 – 11:45	TIChE-AM-23	Development benzoxazine/epoxy composites reinforced with Zylon™ and coir fibers: Effect of stacking sequences	Mr. Prem Sangsenil Srinakharinwirot University, Thailand
11:15 – 11:30	TIChE-AM-24	Fabrication and characterization of PLA/cellulose/biochar composite films for agricultural applications	Ms. Patarapun Wongmanee Kasetsart University, Thailand
11:45 – 12:00	Lunch		9 Dusit 1, 2, 3 (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 23 : Sustainable and Green Chemistry (SG)			📍 Dusit 11 (5 th Floor) 🕒 10:30 – 11:45
Chair : Asst. Prof. Dr. Preuk Tangpromphan Department of Chemical Engineering, Faculty of Engineering, Kasetsart University, Thailand Co-Chair : Assoc. Prof. Dr. Channarong Puchongkawarin Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand			
Time	Abstract Code	Title	Presenter
10:30 – 10:45	TIChE-SG-39	Deacidification of waste cooking oil by adsorption using activated carbon derived from juvenile durian fruit	Ms. Chariyaporn Thanyathippipat King Mongkut's Institute of Technology Ladkrabang, Thailand
10:45 – 11:00	TIChE-SG-40	Hansen solubility parameter-guided selection of alternative solvents for improved extraction of <i>Centella asiatica</i>	Ms. Nuttida Rosmontri Burapha University, Thailand
11:00 – 11:15	TIChE-SG-43	Lithium separation from multicomponent metal solutions using dopamine/polyethyleneimine-modified polyethersulfone membranes	Ms. Janisata Songdaung King Mongkut's University of Technology Thonburi, Thailand
11:15 – 11:30	TIChE-SG-06	Enhancing the esterification yield of iso-amyl acetate from isoamyl alcohol and acetic acid via reactive distillation technique	Mr. Chanatip Somsrang Suranaree University of Technology, Thailand
11:30 – 11:45	TIChE-SG-33	Application of encapsulated phase change material particles in decorative paints for enhanced thermal reduction	Ms. Sangtawan Sappajak Chulalongkorn University, Thailand
11:45 – 12:00	Lunch		📍 Dusit 1, 2, 3 (5 th Floor)
Session 25 : Advanced Materials and Nanotechnology (AM) Process Engineering and Digital Technologies (PE)			📍 Dusit 4 (5 th Floor) 🕒 13:00 – 13:45
Chair : Asst. Prof. Dr. Pawin Lamprasertkun School of Bio-Chemical Engineering and Technology, Sirindhorn International Institute of Technology, Thammasat University, Thailand Co-Chair : Asst. Prof. Dr. Santamon Luanwuthi Department of Industrial Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:15	 TIChE-PE-22	A multi-criteria evaluation framework for selecting parameter ranking methods for CO ₂ balancing in manufacturing	Mr. Ijlal Raheem King Mongkut's University of Technology North Bangkok, Thailand
13:15 – 13:30	TIChE-AM-25	Self-healing composites based on graphene filled bio-based benzoxazine/epoxy nanocomposite reinforced with glass fiber	Mr. Srianon Sangboonrueng Srinakharinwirot University, Thailand
13:30 – 13:45	TIChE-AM-26	Enhancement of electrochemical CO ₂ capture via redox-mediated anthraquinone on Nitrogen-doped porous carbon electrodes	Mr. Jakrin Nawakijwatana Sirindhorn International Institute of Technology (SIIT), Thammasat University, Thailand
13:45 – 14:00	No Event		
14:00 – 14:15	No Event		
14:15 – 14:30	Networking Break (Coffee Break)		📍 Refreshment Zone (5 th Floor)

ORAL PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Day 2 : Wednesday, June 10, 2026

Session 26 : Catalysis and Reaction Engineering (CR)			9 Dusat 5 (5 th Floor) 13:00 – 14:00
Chair : Asst. Prof. Puttaporn Saengthian Department of Chemical Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand Co-Chair : Thai Sangthien Department of Industrial Engineering, Faculty of Engineering, Ubon Ratchathani University, Thailand			
Time	Abstract Code	Title	Presenter
13:00 – 13:15	TIChE-CR-14	Enhanced hydrogen production from water splitting over Zinc Indium Sulfide photocatalysts under visible light	Ms. Patpicha Dolsirittigul Kasetsart University, Thailand
13:15 – 13:30	TIChE-CR-07	Role of magnesium-modified nickel aluminate catalyst for dry reforming of methane	Ms. Oraya Leelaphuthipong Kasetsart University, Thailand
13:30 – 13:45	TIChE-CR-30	PET-derived carbon quantum dots/black TiO ₂ composites for enhanced photocatalytic CO ₂ reduction	Dr. Somchate Wasantwisut Mahidol University, Thailand
13:45 – 14:00	TIChE-CR-33	Reduction of egg white viscosity via enzymatic hydrolysis	Mr. Sirawitch Janapiraganit King Mongkut's Institute of Technology Ladkrabang, Thailand
14:00 – 14:30	Networking Break (Coffee Break)		9 Refreshment Zone (5 th Floor)

END OF ORAL SESSION

POSTER PRESENTATION PROGRAM

The 35th Thai Institute of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Wednesday, June 10, 2026

Poster Session: Advanced Materials and Nanotechnology (AM)		📍 Activity Hall (5th Floor) 🕒 13:00 – 15:00
Abstract Code	Title	Presenter
TIChE-AM-03	Production of cellulose nanofibers from aloe vera peel waste via chemical pretreatment and acid hydrolysis	Assoc. Prof. Dr. Somnuk Jarudilokkul King Mongkut's University of Technology Thonburi, Thailand
TIChE-AM-05	Development of carbon nanotube-based biosensors for detection of biomarkers of lung cancer	Ms. Natcha Tanasukasem Mahidol University, Thailand
TIChE-AM-06	Hyaluronate- α -arbutin conjugates for improved stability of α -arbutin by preventing hydrolytic degradation	Ms. Sawitra Chrannit Chulalongkorn University, Thailand
TIChE-AM-07	Undoped and Zn-doped BiVO ₄ photoanodes for photoelectrochemical oxidation of glycerol	Mr. Natthaphum Daowthaisong Chulalongkorn University, Thailand
TIChE-AM-12	Effect of surfactants on the microencapsulation of methyl palmitate phase change material in melamine formaldehyde shell	Ms. Labhassiree Tansuthpverawongse Chulalongkorn University, Thailand
TIChE-AM-17	Development of electrochemical sensor for the detection of residual pesticide	Mr. Thananchanai Sapphetphisarn Kasetsart University, Thailand
TIChE-AM-18	PEG-assisted dispersion of carbon quantum dots in PLA for enhanced PLA/CQDs composite film performances	Ms. Ausanee Sangrujee Chulalongkorn University, Thailand

Poster Session: Biochemical and Environmental Engineering (BE)		📍 Activity Hall (5th Floor) 🕒 13:00 – 15:00
Abstract Code	Title	Presenter
TIChE-BE-01	Improving photo-Fenton rhodamine B degradation using iron mill scale/graphitic carbon nitride composite	Ms. Nantha Sedtinwong Chulalongkorn University, Thailand
TIChE-BE-02	Syngas wet-scrubbing wastewater treatment using pilot-scale constructed wetlands system	Ms. Niramai Traiyawong Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-BE-04	Influence of calcium ion crosslinking on bentonite-carboxymethylcellulose composites for bromelain immobilization	Ms. Kanlayanit Pimcharoen Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-BE-06	Study of sterilizing-grade filters for producing high dextrose parenteral nutrition	Assoc. Prof. Dr. Asawin Meechai King Mongkut's University of Technology Thonburi, Thailand
TIChE-BE-11	Influence of reaction temperature and reactant molar ratio on the sulfonation of methyl ester to produce α -methyl ester sulfonate using chlorosulfonic acid	Mr. Wanchana Sisuthog Thailand Institute of Scientific and Technological Research (TISTR), Thailand

Abstract Code	Title	Presenter
TIChE-BE-16	Extraction of bioactive compounds from the red macroalga <i>Botryocladia</i> sp.: An understudied marine resource from Thailand	Ms. Rawisara Wisetcholartarn Chulalongkorn University, Thailand
TIChE-BE-17	High-pressure ethanolic extraction of xanthenes from mangosteen pericarp	Assoc. Prof. Dr. Ampai Chanachai King Mongkut's University of Technology Thonburi, Thailand
TIChE-BE-19	Establishing a genome-scale modeling platform for rice callus as a cell factory: Application to α -amylase production	Mr. Chanaphat Pinchaimool King Mongkut's University of Technology Thonburi, Thailand
TIChE-BE-22	Synthesis of mesoporous silica gel from waste solar panels using a gelatin template	Asst. Prof. Dr. Srisuda Thammaratdumrong Burapha University, Thailand

Poster Session: Catalysis and Reaction Engineering (CR)

📍 **Activity Hall (5th Floor)**
🕒 **13:00 – 15:00**

Abstract Code	Title	Presenter
TIChE-CR-03	Leaching of fly ash with amino acid for indirect carbonate mineralization	Mr. Peranut Klayklung Chulalongkorn University, Thailand
TIChE-CR-05	Catalytic methane decomposition for carbon nanotube production using bimetallic catalysts supported on natural aluminosilicate	Ms. Kanoklada Thongdaeng Chulalongkorn University, Thailand
TIChE-CR-08	Green synthesis of titanium dioxide using <i>Capsicum annuum</i> L. extract for photocatalytic degradation of Congo Red dye	Ms. Chayada Phothiwat Chulalongkorn University, Thailand
TIChE-CR-10	Study of direct CO ₂ hydrogenation to dimethyl ether over Cu/ZnO/Al ₂ O ₃ – ZSM-5 catalyst	Mr. Channarong Wajasat Chulalongkorn University, Thailand
TIChE-CR-13	Highly efficient photocatalytic non-oxidative coupling of methane over a novel Ag-modified TiO ₂ –SiO ₂ catalyst	Mr. Natthapong Pinitkajorndech Kasetsart University, Thailand
TIChE-CR-17	Influence of Si/Al ratio of ZSM-22 on hydroconversion of palm kernel oil over Ni-based catalysts for bio-naphtha and bio-jet fuel production	Ms. Artitaya Chullakran Chulalongkorn University, Thailand
TIChE-CR-18	Hydrogen production from waste plastic pyrolysis oil via two-stage catalytic gasification integrated with water-gas shift reaction	Ms. Parichad Bunma Chulalongkorn University, Thailand
TIChE-CR-23	Hydrophobic surface modification of Cu/ZnO catalysts for enhanced methanol synthesis in ethanol-assisted CO ₂ hydrogenation	Mr. Suvapit Kulphanan Chulalongkorn University, Thailand
TIChE-CR-25	Study of oxygen adsorption on PtPd-based high-entropy-alloy catalysts via density functional theory and machine learning	Mr. Potipak Tantitumrongwut Chulalongkorn University, Thailand
TIChE-CR-29	Development of a kinetic model and process simulation for diethyl ether production from ethanol dehydration over unmodified H-form beta zeolite catalyst	Mr. Chutiman Chaleawphudphong Chulalongkorn University, Thailand
TIChE-CR-35	Zr-based UiO-68 MOFs functionalized with piperazine for efficient cycloaddition of CO ₂ with propylene oxide	Ms. Aimwipa Sasiprapa Kasetsart University, Thailand

Abstract Code	Title	Presenter
TIChE-CR-36	Design and scale-up of a continuous bubble slurry reactor for CO ₂ absorption with MEA and CaCO ₃ mineralization using high-calcium coal fly ash	Mr. Marucheth Thongthepairoj Chulalongkorn University, Thailand
TIChE-CR-38	Electrodeposition of Ag on Ti felt for morphology-controlled catalyst design	Mr. Montree Obchoei King Mongkut's University of Technology Thonburi, Thailand
TIChE-CR-39	Enhancing CO ₂ hydrogenation over Co/γ-Al ₂ O ₃ Catalysts via ZrO ₂ promotion and base modification	Ms. Kanjanaphon Dedpon King Mongkut's University of Technology North Bangkok, Thailand

Poster Session: Energy and Fuels (EF)

📍 Activity Hall (5th Floor)
🕒 13:00 – 15:00

Abstract Code	Title	Presenter
TIChE-EF-02	Bio-based demulsifier from agricultural waste: TEOS-modified rice husk and lotus leaf for efficient water-in-crude oil emulsion separation	Assoc. Prof. Dr. Somnuk Jarudilokkul King Mongkut's University of Technology Thonburi, Thailand
TIChE-EF-03	Mitigation of slagging tendency in 3-stage gasification biochar from Thai agricultural residues via water and acid leaching Pretreatment	Mr. Chatuphon Uthaisri Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-EF-04	Application of degraded biodiesel as a tar scrubbing medium in venturi wet scrubbing for biomass gasification system	Ms. Thanita Sonthisawate Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-EF-10	Engineering evaluation of biomass stove design parameters and fuel types for sustainable community-scale energy applications	Mr. Tanakorn Varapiang Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-EF-15	Optimization of sugar extraction from sugarcane leaves via acid-enzymatic hydrolysis for biohydrogen production	Ms. Angkana Khuenpetch Thailand Institute of Scientific and Technological Research (TISTR), Thailand

Poster Session: Industrial Applications and Case Studies (IA)

📍 Activity Hall (5th Floor)
🕒 13:00 – 15:00

Abstract Code	Title	Presenter
TIChE-IA-05	Pilot-scale assessment of aqueous absorbent for CO ₂ capture	Ms. Rattanaorn Apaiyakul Chulalongkorn University, Thailand

Poster Session: Process Engineering and Digital Technologies (PE)

📍 Activity Hall (5th Floor)
🕒 13:00 – 15:00

Abstract Code	Title	Presenter
TIChE-PE-07	Techno-economic analysis of integrated flue gas pretreatment and CO ₂ capture from oil refinery using Aspen Plus	Mr. Sirawit Srithammavong Chulalongkorn University, Thailand
TIChE-PE-15	Techno-economic and carbon footprint analysis of hydrogen production from glycerol via supercritical water reforming	Ms. Jintana jumpanui Silpakorn University, Thailand

Poster Session: Sustainable and Green Chemistry (SG)		9 Activity Hall (5 th Floor) 13:00 – 15:00
Abstract Code	Title	Presenter
TIChE-SG-03	Valorization of sugarcane filter cake into biochar as a cement additive	Ms. Thanita Sedtinwong Concord College, England
TIChE-SG-04	Regeneration of CO ₂ -rich monoethanolamine absorbent using HZSM-5 catalyst in fluidized stripper	Ms. Kanyarat Saechua Chulalongkorn University, Thailand
TIChE-SG-07	Dynamic pH Control in MEA-Based CO ₂ Mineralization: A Comparison of CaCl ₂ and Ca(OH) ₂	Ms. Unchisa Boonkun Chulalongkorn University, Thailand
TIChE-SG-10	Influence of CO ₂ -rich biogas reforming syngas on methanol production, carbon utilization, and catalyst stability over Cu/ZnO/Al ₂ O ₃	Dr. Rujira Jitrwung Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-SG-13	Synthesis of bimetallic Ni-Fe catalysts for carbon nanotube synthesis via CO ₂ electrolysis	Mr. Pathompat Khowattana Silpakorn University, Thailand
TIChE-SG-14	Effect of deep eutectic solvent type on the recycling of polyester/cotton blended textile waste using a DES–NaOH hybrid system	Ms. Piyathida Wangaopklang Rajamangala University of Technology Thanyaburi, Thailand
TIChE-SG-15	Utilization of zeolite-A obtained from different biomass sources as support of zirconocene/MAO catalysts for ethylene polymerization	Mr. Pakornkeat Okapanom Chulalongkorn University, Thailand
TIChE-SG-16	Absorption and regeneration performance of MEA–Water–EG absorbent for carbon capture	Mr. Nattayot Photim Chulalongkorn University, Thailand
TIChE-SG-18	Recycling zinc-carbon batteries into Na _x MnO ₂ cathode materials for sodium-ion batteries	Ms. Panisara Kerdudom Chulalongkorn University, Thailand
TIChE-SG-20	Techno-economic evaluation of ethanol-assisted CO ₂ hydrogenation to methanol under different reactor configurations	Mr. Kritchanon Sriwilai Chulalongkorn University, Thailand
TIChE-SG-21	Electrochemical synthesis of alkane from fatty acid in a microreactor	Ms. Angkhana Saeoueng Chulalongkorn University, Thailand
TIChE-SG-22	Scaling-up of biolubricant production from palm oil fatty acid methyl ester via transesterification with trimethylolpropane	Mr. Sanit Luek-aram Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-SG-23	Palm trunk-derived hydrochar as a sustainable catalyst support for trimethylolpropane triester synthesis	Ms. Pimchanok Longchin Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-SG-27	Pyrolysis oil from empty fruit bunches: Alternative resources for sustainable fuels and chemicals	Mr. Tosaporn Phetyim Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-SG-28	Urea-assisted fractionation of FAME: Impact on fatty acid composition and cold flow behavior	Ms. Punyaporn Khamdaeng Thailand Institute of Scientific and Technological Research (TISTR), Thailand
TIChE-SG-29	Using alkali carbonate and amine-based modified sugarcane bagasse ash as a low-cost solid sorbent for post-combustion CO ₂ capture	Ms. Natchar Saesue Mahidol University, Thailand

Abstract Code	Title	Presenter
TIChE-SG-32	Field implementation and performance evaluation of palm-oil-based bio-transformer oil (EnPAT) in distribution transformers	Mr. Bowornchai Chareonteraboon National Science and Technology Development Agency, Thailand
TIChE-SG-34	Synthesis of high-stability biotransformer oil from palm derivatives for open-type transformer applications	Dr. Sareena Mhadmhan National Science and Technology Development Agency, Thailand
TIChE-SG-35	Recovery of vitexin and iso-vitexin from mung bean seed coat waste by ethanolic percolation extraction and investigation of the effects of storage time and temperature on liquid extract degradation	Ms. Nuntapohn Chanthai Kasetsart University, Thailand
TIChE-SG-38	Synthesis of trimetallic Ni-Fe-Co catalysts for carbon nanotubes via through CO ₂ molten electrolysis	Asst. Prof. Dr. Natthawan Prasongthum Khon Kaen University, Thailand
TIChE-SG-41	Life cycle assessment of coffee process in an agroforestry system: A case study from Ban Pa Miang, Lampang, Thailand	Ms. Tubtim Pattharathanon King Mongkut's University of Technology Thonburi, Thailand
END OF POSTER SESSION		

TIChE SENIOR PROJECT 2026 PROGRAM
The 35th Thai Institution of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Tuesday, June 9, 2026

Session : TIChE Senior Project Contest 2026 (Applied Research)		📍 Activity Hall (5 th Floor) 🕒 09:00 – 16:30
Project Code	Title	Contestant
TIChE-SE-A01 MJU	Synthesis of biochar from corn husk functionalized with silver and iron oxides for the removal of organic pollutants and antibacterial activity against Escherichia coli	(1) Chanyanut Thammasit (2) Asst. Prof. Dr. Sakdinun Nuntang Department of Industrial Chemistry Innovation, Maejo University
TIChE-SE-A02 CU	Modification of ZSM-5 via dealumination for producing bio-naphtha derived from hydroconversion of biodiesel over Ni-Co catalysts	(1) Thaiyada Rinampha (2) Pongsakorn Kaewsuwan (3) Prof. Dr. Napida Hinchiranan Department of Chemical Technology, Chulalongkorn University
TIChE-SE-A03 SWU	Development of sustainable wood coatings for the preservation of teak structures at Mrigadayavan palace	(1) Napatchakorn Saipim (2) Jinpanee Akaramahaasenabordee (3) Dr.Lunjakorn Amornkitbamrung" Department of Chemical Engineering, Srinakharinwirot University
TIChE-SE-A04 PSU	Development of soil reinforcement materials from natural rubber	(1) Pornnapa Wongngam (2) Pannaray Putkhaw (3) Asst. Prof. Dr. Pornsiri Kaewpradit Department of Chemical Engineering, Prince of Songkla University
TIChE-SE-A05 MU	Fabrication of hydrophobic wood by in situ polymerization of furfuryl alcohol resin for furniture application	(1) Kanyanut Siriwatanarat (2) Isaree Lertsutthiphol (3) Assoc. Prof. Dr. Atthapon Srifa (4) Asst. Prof. Dr. Wanida Koo-amornpattana Department of Chemical Engineering, Mahidol University
TIChE-SE-A06 KU	Ammonia-evaporated Cu/SiO ₂ catalysts for low-temperature CO ₂ -to-CO upgrading with kinetic analysis toward reactor design	(1) Benyaporn Thamaputikul (2) Phetmamnueng Kaeoruegrit (3) Prof. Dr. Thongthai Wittoon Department of Chemical Engineering, Kasetsart University
TIChE-SE-A07 SUT	Craft beer production: experimental, process simulation, and economic analysis	(1) Thanapat Srimart (2) Jirawut Phloedchanhuek (3) Assoc. Prof. Dr. Lek Wantha School of Chemical Engineering, Suranaree Univeristy of Technology
TIChE-SE-A08 KKU	Fabrication and characterization of microneedle patches incorporating alpha-mangostin and aloe vera extract	(1) Nattapong Suthongsa (2) Saithip Mulnoisu (3) Assoc. Prof. Dr. Duangkanok Tanangteerapong Department of Chemical Engineering, Khon Kaen University

Project Code	Title	Contestant
TIChE-SE-A09 BUU	Web-based machine learning models for real-time cavitation prediction using remote smartphone accelerometer	(1) Anucha Khumphokha (2) Duangruedi Maikonklang (3) Patiparn Boonruam Department of Chemical Engineering, Burapha University
TIChE-SE-A10 KMUTT	CFD simulation of hydrodynamics in an internally circulating fluidized bed for direct air CO ₂ capture	(1) Amarawadee Charuensuk (2) Rungnapha Pongchim (3) Dr. Krittin Korkerd Department of Chemical Engineering, King Mongkut's University of Technology Thonburi
TIChE-SE-A11 CU	Develop smelling test kit for Parkinson disease screening by using encapsulation technology	(1) Phurinat Priampiyanusorn (2) Nannapat Ratanasirivilai (3) Assoc. Prof. Dr. Apinan Soottitantawat (4) Asst. Prof. Dr. Rungthiwa Methaapanon Department of Chemical Engineering, Chulalongkorn University
TIChE-SE-A12 RMUTT	Synthesis of hard carbon from coconut shell for use as an anode electrode	(1) Wiphawanee Chuchan (2) Sasithorn Sukpheuchkit (3) Assoc. Prof. Dr. Chaiyan Chaiya Department of Chemical and Materials Engineering, Rajamangala University of Technology Thanyaburi
TIChE-SE-A13 SU	Process optimization and economic valorization of sulphur recovery from sludge generated in biogas H ₂ S removal systems via solvent extraction	(1) Kanlayanee Pansap (2) Thansiri Phiongam (3) Asst. Prof. Dr. Weerawat Patthaveekongka Department of Chemical Engineering, Silpakorn University
TIChE-SE-A14 UBU	Optimum conditions for enhancing the softness and strength of tissue pulp from rice straw	(1) Kanokwan Saisavat (2) Asst. Prof. Dr. Tiammanee Rattanaweeran (3) Asst. Prof. Dr. Idtisak Paopo Department of Chemical and Environmental Engineering, Ubon Ratchathani University
TIChE-SE-A15 TU	CO ₂ Utilization of the final stage Li-S leachate	(1) Saharat Sutthitham (2) Phakwalan Prompim (3) Asst. Prof. Dr. Woraratana Pattaraprakorn Department of Chemical Engineering, Thammasat University

**Session : TIChE Senior Project Contest 2026
(Basic Research)**

📍 Activity Hall (5th Floor)
🕒 09:00 – 16:30

Project Code	Title	Contestant
TIChE-SE-B01 SWU	Effect of Vernonia amygdalina extract in jellyfish gelatin coated on polylactic acid (PLA)/ Poly(butylene adipate-co-terephthalate) (PBAT) for wound dressing application	(1) Rattanaprapha Wannasiri (2) Natteera Yuchuakan (3) Assoc. Prof. Dr. Pollawat Charoeythornkhajhornchai Department of Chemical Engineering, Srinakharinwirot University
TIChE-SE-B02 CU	Designing of direct air carbon dioxide capture prototype using monolith structures	(1) Nutnicha Wisetsakorn (2) Oranan Worawetthanakul (3) Prof. Dr. Benjapon Chalermisinsuwan Department of Chemical Technology, Chulalongkorn University

TIChE-SE-B03 PSU	Morphology and distribution characteristics of airborne microplastics in an urban atmosphere	(1) Kamarulzaman Liming (2) Isariya Silasalaisophit (3) Assoc. Prof. Dr. Racha Dejchanchaiwong Department of Chemical Engineering, Prince of Songkla University
TIChE-SE-B04 MU	Machine learning model for competitive CO ₂ and N ₂ separation from industrial flue gas via metal-organic frameworks	(1) Panida Somduangsri (2) Sitthanan Pongcharoen (3) Asst. Prof. Dr. Poomiwat Phadungbut Department of Chemical Engineering, Mahidol University
TIChE-SE-B05 KU	Molecular docking of curcumin derivatives as drug against colorectal cancer	(1) Phatthanant Sutthae (2) Suwichaya Suasamserm (3) Dr. Kulpavee Jitapunkul Department of Chemical Engineering, Kasetsart University
TIChE-SE-B06 SUT	Study and development of new chemical formulas for cleaning artificial knee joints	(1) Chumphon Rotchomphoo (2) Dr. Supunnee Janpirom School of Chemical Engineering, Suranaree University of Technology
TIChE-SE-B07 KKU	Residue coffee extraction with CNC - PVA film for UV-blocking and antioxidant properties	(1) Weerachon Jermungnorn (2) Putcharapong Kongtaworn (3) Assoc. Prof. Dr. Khanita Kamwilaisak Department of Chemical Engineering, Khon Kaen University
TIChE-SE-B08 KMUTT	The effect of feed flow rate on membrane bioreactor performance for rubber wastewater treatment	(1) Haruthai Somboonthana (2) Thinnakorn Woraseeha (3) Asst. Prof. Dr. Waritha Jantaporn (4) Dr. Kiattinatapon Juengchareonpoon Department of Chemical Engineering, King Mongkut's University of Technology Thonburi
TIChE-SE-B09 CU	Large-scale microreactor for electrochemical conversion of fatty acids to alkanes	1) Phawat Tantiseri 2) Napat Jirapawasut 3) Prof. Dr. Varong Pavarajarn Department of Chemical Engineering, Chulalongkorn University
TIChE-SE-B10 UBU	Cultivation of <i>Scenedesmus</i> sp. microalgae in a bubble column photobioreactor	(1) Kittithon Hemangkur (2) Nawaphon Youkong (3) Asst. Dr. Idtisak Paopo (4) Assoc. Prof. Dr. Chatchai Kunyawut Department of Chemical and Environmental Engineering, Ubon Ratchathani University
TIChE-SE-B11 TU	PID controller tuning for a continuous stirred-tank reactor (CSTR) using Aspen HYSYS V12.1	(1) Chirasuta Chitchong (2) Thanathrid Wongchotsathit (3) Assoc. Prof. Dr. Satok Chaikunchuensakun Department of Chemical Engineering, Thammasat University

END OF TIChE SENIOR PROJECT CONTEST 2026 SESSION

TIChE HIGH SCHOOL PROJECT 2026 PROGRAM
The 35th Thai Institution of Chemical Engineering and Applied Chemistry International Conference (TIChE2026)

Wednesday, June 10, 2026

Session : TIChE High School Project Contest 2026 (Poster Competition)		📍 Activity Hall (5th Floor) 🕒 09:00 – 12:00
Project Code	Title	Contestant
TIChE-HS-01	Development of a Sunscreen Stick from Peel Extracts of Kaew Khamin Mango (<i>Mangifera indica</i> L.)	(1) Laksamon Kongjalern (2) Natjirapa Hongbin (3) Waralee Nopsri (4) Suphap Paendee (5) Pattamon Viriyatham Chonkanyanukoon, Chonburi
TIChE-HS-02	Determination of Copper and Lead ions in Water Using Filter Paper Modified with Reagent Extracted from Beetroot mixed with Chitosan	(1) Chetnutch Khiewplueang (2) Natthaphak Assawapatchara (3) Naphassakorn Fongfung Phichai Rattanakhan, Ranong
TIChE-HS-03	Development of Composite Materials from Bacterial Cellulose Incorporated with Anthocyanin and ZIF-8 Metal-Organic Framework for Monitoring the Freshness of Raw Meat	(1) Chetnipit Boonchareamrat (2) Natdanai Siengpaikro (3) Peem Pakkruek (4) Manas Sittishoktram Princess Chulabhorn Science High School Phetchaburi, Phetchaburi
TIChE-HS-04	Optimization of Ethanol Concentration and Comparative Evaluation of Extraction Techniques for Polyphenol Recovery from Areca seed (<i>Areca catechu</i> L.)	(1) Nathakorn Thurapan (2) Khanti Buttanadamrongchit (3) Puritat Sripanya (4) Ploypailin Ponsongkhor KhonkaenWittayayon Khon Kaen
TIChE-HS-05	Effects of Cold Plasma on Citronella and Turmeric Essential Oil Extracts and Neem Crude Extract in Controlling of Rice Weevils (<i>Sitophilus oryzae</i>)	(1) Sorawit Sonthip (2) Phachara Panyaphet (3) Thalit vaivang (4) Supanut Pattanasarin (5) Kannika Pongkan Piriyalai School Phrae, Phrae
TIChE-HS-06	Effect of Cold Plasma on the Antioxidant Activity of Essential Oils from Citronella and Turmeric	(1) Peerada Tippayawong (2) Artittaya Kuekkong (3) Tamonwan U-parnanchaichana (4) Supanut Pattanasarin (5) Kannika Pongkan Piriyalai School Phrae, Phrae
TIChE-HS-08	Privacy-Preserving Federated Learning Framework for Distributed Chemical Process Optimization	(1) Teetat Pipattaratonthai (2) Aueaphum Aueawattthanaphisut Triam Udom Suksa School, Bangkok
TIChE-HS-09	Study of the Properties and Performance of Silica Extracted from Rice Husk	(1) Ananyanan Tailatbamrung (2) Naphat Chuwannamas (3) Sadanan Samphan (4) Kodchakorn Lamsombat (5) Egkachai Srisupap Kanlayanawat School, Khon Kaen
TIChE-HS-10	Closing the Loop with Tannins: Innovative Circular Solutions for Biodegradable Antioxidant Film Production	(1) Worakamol Charearnsom (2) Suradet Sritha Kasetsart University Laboratory School, Center for Educational Research and Development, Bangkok

END OF TIChE HIGH SCHOOL PROJECT CONTEST 2026 SESSION